

DEPT: DECOUPLED EMBEDDINGS FOR PRE-TRAINING LANGUAGE MODELS

Alex Jacob, Lorenzo Sani, Meghdad Kurmanji, William F. Shen, Xinchu Qiu, Dongqi Cai, Yan Gao, Nicholas D. Lane

University of Cambridge, Flower Labs, Beijing University of Posts and Telecommunications

May 24, 2026

Why heterogeneous pre-training

- Large language models power search, assistants, coding, and translation at global scale.
- Pre-training on diverse, massive corpora drives generalization and downstream success.
- Heterogeneous data (domains, languages) is essential but creates efficiency and stability challenges.

Interference and inefficiency

- Negative interference: domains/languages compete for limited capacity.
- Curse of multilinguality: diminishing returns as languages increase, low-resource hurt most.
- Embedding bottleneck: shared giant vocabularies dilute tokens and dominate parameters (40–80%).

Why existing practice struggles

- Heavy HPO: sampling temperatures per model–tokenizer pair; brittle stability [Arivazhagan et al., 2019].
- One-size-fits-all tokenizers: subword fragmentation and fertility across languages.
- Distributed training costs: embeddings dominate memory and communication in multilingual setups.

- Train one transformer body across heterogeneous sources without a shared vocabulary.
- Keep token/positional embeddings local to each source; only aggregate the body.
- Federated/meta-learning style: local InnerOPT, global OuterOPT (FedAvg-like regularizer) [McMahan et al., 2017].



- Offline: build tokenizers (global or per-source), pre-tokenize each source.
- Inner loop: local training per source for a fixed number of local steps (AdamW; Loshchilov & Hutter, 2019).
- Outer loop: aggregate transformer bodies; embeddings handled per-variant.



- GLOB: shared global embeddings; local training reduces comms by syncing every N_{local} steps.
- TRIM: per-source trimmed embeddings mapped to global; shrink local memory and comms.
- SPEC: fully decoupled token and positional embeddings per source; exchange only the body.

- Body averaging injects stabilizing noise.
- Lowers activation and parameter norms.
- Enables stable training under fixed hyperparameters.



- TRIM: select source rows from the global embeddings; expand back by zero-padding before aggregation.
- SPEC: never aggregates token or position embeddings; only the transformer body is averaged.
- Positional specialization in SPEC supports unknown or private sequence characteristics.

- Savings increase with more local steps between aggregations.
- Larger gap between global and per-source vocabulary sizes yields larger savings.
- SPEC minimizes communication by not transmitting embeddings at all.

- SPEC-OPT cuts per-step communication by about 1600x at a 12-block multilingual setting.
- Embedding memory shrinks from 192M to 38.6M (about 5x) for SPEC-OPT.
- At 1B scale: 714x communication reduction; 24% model memory reduction.

See Table 2 below.

Table: Table 2: Practical memory and communication costs for DEPT

Type	# Blocks	Method	n_{local}	T	$V_b \times d_v$	$V_b \times d_{model}$	M_b (M)	Per-step Comm Cost (M)	
Multilingual	12	STD	5×103	1	250 112	192M	278 M (1x)	278 M (1x)	
Multilingual	12	GLOB	500	10	250 112	192M	278 M (1x)	0.26 M (0.000x)	
Multilingual	12	TRIM	500	10	216 135 $\pm$ 27 160	166M	252 M (0.90x)	0.5 M (0.002x)	
Multilingual	12	SPEC	500	10	216 135 $\pm$ 27 160	166M	252 M (0.90x)	0.17 M (0.0006x)	
Multilingual	12	SPEC-OPT	500	10	50 257 $\pm$ 0	38.6M	125 M (0.45x)	0.17 M (0.0006x)	
Multilingual (1B)	24	STD	7×103	1	250 112	512.2M	1.71 B (1x)	1.71 B (1x)	
Multilingual (1B)	24	SPEC-OPT	500	14	50 257 $\pm$ 0	102.9M	1.3B (0.76x)	2.4 M (0.001x)	

- Models: decoder-only transformers, 125M–1.3B parameters.
- Data: The Pile (16 subsets) and MC4 (8 langs), plus OOD splits.
- Metrics: validation perplexity (in/OOD), downstream tasks; stability via norms; plasticity via adaptation speed.

- Pile (24-block): about +15.3% average perplexity improvement; 10–20% OOD gains.
- MC4 (12-block): about +17.3% average in-distribution; up to +20.8% OOD.
- OuterOPT reduces norms and prevents divergence under heterogeneity.



- Average improvement of about 10–15% vs STD across subsets.
- SPEC-OPT is strongest on high UNIGRAM-CE subsets.
- Robust to embedding re-initialization across runs.

- Average in-distribution improvement of about 14–17%; OOD gains up to 20.8%.
- Consistent benefits across languages with varied tokenization difficulty.
- SPEC tends to be strongest on high-entropy OOD languages.

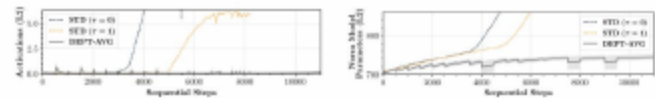
- DEPT improves MNLI, RACE, STSB, SST-2 by about 3–7.5% relative.
- TRIM is often strongest, suggesting better embedding–body synergy.
- Gains reflect stronger, more plastic transformer bodies.

Table: Table 7: Downstream task performance (↑) after continued pre-training

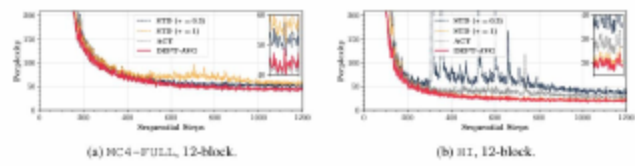
Name	Random Init RACE (ACC)	MNLI (ACC)	STSB (PC)	SST2 (ACC)						
STD ($\tau = 0$)	0.50	0.60	0.66	0.79						
STD ($\tau = 1$)	0.46	0.68	0.73	0.81						
GLOB	0.51	0.72	0.78	0.83						
TRIM	0.53	0.71	0.78	0.83						
SPEC	0.52	0.71	0.79	0.81						
SPEC-OPT	0.51	0.69	0.77	0.85						
Min Imp (%)	2.9%	4.6%	5.9%	-0.7%						
Max Imp (%)	5.8%	6.1%	7.5%	4.1%						

- DEPT adapts faster to new languages and domains from random embeddings.
- Reaches lower final perplexity on full distribution and OOD languages.
- SPEC's decoupled positions help transfer structural priors across sources.

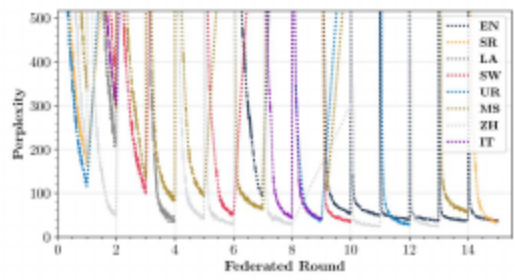
- First demonstration: SPEC converges stably with client subsampling.
- Late-introduced high-resource languages integrate without instability.
- Orders-of-magnitude comms reduction makes cross-silo training practical.



- Rapid convergence on full MC4, low-resource SW, and OOD (HI, DE).
- Lower perplexity floors indicate better cross-lingual transfer.
- Supports training with per-source optimized vocabularies (SPEC-OPT).



- On IID C4, DEPT matches standard perplexity aside from brief OuterOPT spikes.
- Maintains lower activations, suggesting training stability benefits.
- Indicates DEPT is not limited to heterogeneous settings.



- After SPEC training, build a shared embedding via short continued pre-training
- Alternatives: vocabulary matching or stitching; zero-shot mapping is open.
- This post-processing was applied uniformly across all methods.

- Zero-shot/global embedding construction without continued pre-training.
- Adaptive aggregation and client sampling; improved OOD alignment.
- Extend decoupling to more modalities; further scale federated pre-training.

- We thank collaborators and supporting institutions.
- Gratitude to dataset providers and open-source communities.

- Temperature-based multilingual sampling: Arivazhagan, Naveen, et al. 2019. Massively Multilingual Neural Machine Translation in the Wild.
- FedAvg (OuterOPT): McMahan, H. Brendan, et al. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data.
- AdamW (InnerOPT): Loshchilov, Ilya, and Frank Hutter. 2019. Decoupled Weight Decay Regularization.
- Implementation refs (optional): Kudo, T. 2018. SentencePiece; Press, O., et al. 2022. ALiBi.

- Can we build a global embedding without continued pre-training?
- What aggregation/client sampling best stabilizes small models?
- How to align OOD/private embeddings without leakage?
- Can decoupling extend to speech/vision tokenizers at scale?